Equation

$\Rightarrow 0$

0

$\Rightarrow 0$

Note :- $a^{(\log_a x)^2} \Rightarrow x^{\log_a x}$

x Can be only "1" not "-1" Cause $x \geq 0$.

$\therefore$ Sum of all Sol^n $\Rightarrow 1$.

Ex) Find the value of x : is 9 and $\frac{1}{9}$

$\log_3^2 x + x^{\log_3 x} \Rightarrow 162$

$\Rightarrow 3^{\log_3^2 x} + x^{\log_3 x} \Rightarrow 162$

$\rightarrow \left(3^{(\log_3 x)\log_3 x}\right) + x^{\log_3 x} \Rightarrow 162$

$\rightarrow \left(x^{(\log_3 3)(\log_3 x)}\right) + x^{\log_3 x} \Rightarrow 162$ ($\log_3 3 \rightarrow 1$)

$\rightarrow x^{\log_3 x} + x^{\log_3 x} \Rightarrow 162$

$\rightarrow 2\, x^{\log_3 x} \Rightarrow 162$

$\rightarrow x^{\log_3 x} \Rightarrow 81$

$\rightarrow$ Multiply and divid by $\log_3$

$\rightarrow \log_3 x^{\log_3 x} \Rightarrow \log_3 81$, $3^t \Rightarrow 81$, $3^t \Rightarrow 3^4$, $t \Rightarrow 4$

$\rightarrow \log_3 x \cdot \log_3 x \Rightarrow 4$

$\rightarrow (\log_3 x)^2 \Rightarrow 4$

$\rightarrow \log_3 x \Rightarrow \sqrt{4} \Rightarrow \pm 2$, $\log_3 x \Rightarrow 2$ and $\log_3 x \Rightarrow -2$

$3^2 \Rightarrow x$, $x \Rightarrow 9$ & $x \Rightarrow \frac{1}{9}$

Ex) find the value of x satisfying

$\log_{10}(2^x + x - 41) \Leftrightarrow x(1 - \log_{10} 5)$

$\Rightarrow \log_{10}(2^x + x - 41) \Leftrightarrow x(1 - \log_{10} 5)$

$\Rightarrow \log_{10} 2^x + \log_{10} x - \log_{10} 41 \Leftrightarrow x - x\log_{10} 5$

$\Rightarrow 2\log_{10} 2 + \log_{10} x - \log_{10} 41 \Leftrightarrow x - x\log_{10} 5$

$\Rightarrow 2\log_{10} 2x - \log_{10} 41 \Rightarrow x - x\log_{10} 5$

$\rightarrow 2\log_{10} 2x/41 \Rightarrow x - x\log_{10} 5$ } made mistake here

$\rightarrow 2\log_{10}\left(\frac{2x}{41}\right) + x\log_{10} 5 \Rightarrow x$

$\Rightarrow \log_{10}\left(\frac{2x}{41}\right)^2 + \log_{10} 5^x \Rightarrow x$

$\rightarrow \log_{10} \frac{4x^2}{1681} \times 5^x \Rightarrow x$

$\rightarrow \log_{10} \frac{4x^2 \times 5^x}{1681} \Rightarrow x$

$\rightarrow \frac{4x^2}{1681} \times 5^x \Rightarrow 10^x$

$\rightarrow \frac{4x^2}{1681} \rightarrow \frac{2^x \times 5^x}{5^x}$

$\rightarrow \frac{4x^2}{1681} \Rightarrow 2^x$

(Wrong approach)

$\Rightarrow \log_{10}(2^x + x - 41) \Leftrightarrow x(1 - \log_{10} 5)$

$\Rightarrow \log_{10} 2^x + \log_{10} x - \log_{10} 41 \Leftrightarrow x - x\log_{10} 5$

$\Rightarrow x\log_{10} 2 + \log_{10} x - \log_{10} 41 \Rightarrow x - x\log_{10} 5$

$\Rightarrow x\log_{10} 2 + x\log_{10} 5 + \log_{10} x - \log_{10} 41 - x \Rightarrow 0$

$\Rightarrow x(\log_{10} 2 + \log_{10} 5) \to 1 + \log_{10} x - \log_{10} 41 \Rightarrow x$

$\Rightarrow x + \log_{10} x - \log_{10} 41 \Leftrightarrow x$

$\Rightarrow \log_{10} x/41 \Rightarrow 0$

$\Rightarrow 10^0 \Rightarrow \frac{x}{41}$

$\Rightarrow \underline{x \Rightarrow 41}$

Ex) The Product of all the Sol^n of the Eq^n

$x^{1+\log_{10} x} \Rightarrow \underline{100000x}$ is.

$\Rightarrow x^{1+\log_{10} x} \Rightarrow \underline{100000x}$

$\Rightarrow x^{\log_{10} 10 + \log_{10} x} \Rightarrow \underline{100000x}$

$\Rightarrow x^{\log_{10} 10x} \Rightarrow \underline{100000x}$

$\Rightarrow x \cdot x^{\log_{10} x} \Rightarrow \underline{100000x}$

$\Rightarrow x^{\log_{10} x} \Rightarrow \underline{100000}$

$\Rightarrow x^{\log_{10} x} \Leftrightarrow 100000$

$\log_{10}$ on Both Sides

$\Rightarrow \log_{10} x \cdot \log_{10} x \Rightarrow \log_{10} 100000$

$\Rightarrow (\log_{10} x)^2 \Rightarrow 5$

$\Rightarrow \log_{10} x \Rightarrow \pm\sqrt{5}$

$\Rightarrow x \Rightarrow 10^{+\sqrt{5}}$ and $10^{-\sqrt{5}}$

$\Rightarrow$ Product of the Solution $\Rightarrow 10^{\sqrt{5}} \times 10^{-\sqrt{5}}$

$\Rightarrow \frac{10^{\sqrt{5}}}{10^{\sqrt{5}}} \Rightarrow 1$

Ex) find the value of x:

$(x+1)^{\log_{10}(x+1)} \Rightarrow 100(x+1)$

$\Rightarrow \log (x+1)^{\log_{10}(x+1)} \Leftrightarrow 100(x+1)$

$\Rightarrow$ Multiply $\log_{10}$ on Both sides

$\Rightarrow \log_{10}(x+1)^{\log_{10}(x+1)} \Rightarrow \left(\log_{10} 100(x+1)\right)^{\log_{10} 100x + \log_{10}}$

$\Rightarrow \log_{10}(x+1) \cdot \log_{10}(x+1) \Rightarrow \log_{10} 100x + \log_{10} 100$

$\rightarrow (\log_{10}(x+1))^2 \Rightarrow \log_{10} 10^2 + (\log_{10} x+1)$

$\Rightarrow \log_{10} x+1 = t$

$\rightarrow (t)^2 = 2 + t$

$\Rightarrow t^2 - t - 2 \Rightarrow 0$

$\Rightarrow t^2 - 2t + t - 2 \Rightarrow 0$ $\quad 2t \pm 2$

$\Rightarrow t(t-2) + 1(t-2) \Rightarrow 0$ $\quad 2t \wedge t$

$\rightarrow (t-2)(t+1) \Rightarrow 0$

$t \Rightarrow 2$ & $t \Rightarrow -1$

$\log_{10} x+1 \Rightarrow 2$ & $\log_{10} x+1 \Rightarrow -1$

$x+1 \Rightarrow 100$ and $\frac{1}{10}$

$x \Rightarrow 99 \Rightarrow \frac{1}{10} - 1$

$x \Rightarrow \frac{99}{10}$ and $-\frac{9}{10}$

$x > 0$

$\{x \Rightarrow 99\}$

Ex) find the value of x:

$\log_5(5^{1/x} + 125) \Rightarrow \log_5 6 + 1 + \frac{1}{2x}$

$\rightarrow \log_5 [5^{1/x} + 125] \Rightarrow \log_5 6 + 1 + \frac{1}{2x}$

$\rightarrow \log_5 5^{1/x} + \log_5 125 - \log_5 6 \Rightarrow 1 + \frac{1}{2x}$

$\rightarrow \log_5 5^{1/x} \times 5^3 - \log_5 6 \Rightarrow 1 + \frac{1}{2x}$

$\rightarrow \log_5 \frac{5^{\frac{1}{x}+3}}{6} \Rightarrow \frac{2x+1}{2x}$

$\rightarrow \frac{\frac{1}{x}+3}{6} \Rightarrow 5^1 \times 5^{\frac{1}{2x}}$

$\rightarrow \frac{(5^{1/x}) + 5^3}{6} \Rightarrow 5^1 \times (5^{\frac{1}{x}}(2))$

$\rightarrow$

$\rightarrow \Rightarrow 5 \times$

$\rightarrow \Rightarrow 5 \times$

$\rightarrow$ Let $5^{1/2x} \Rightarrow t$

$(5^{\frac{1}{2x}})^2 \Rightarrow t^2$

$t^2 \Rightarrow 5^{1/x}$

$1 + \frac{1}{2x}$

$+1 + \frac{1}{2x}$

$1 + \frac{1}{2x}$

$\Rightarrow \frac{t^2 \times 125}{6} \Rightarrow 5t$

$\Rightarrow \frac{t^2 + 125}{6} - 5t \Rightarrow 0$

$\Rightarrow t^2 - 30t + 125$

$\Rightarrow t \Rightarrow 25$ and 5

$t \Rightarrow 25$ and $t \Rightarrow 5$

$5^{\frac{1}{2x}} \Rightarrow 5^2$ and $5^{\frac{1}{2x}} \Rightarrow 5$

$\frac{1}{2x} \Rightarrow 2$ $\quad \frac{1}{2x} \Rightarrow 1$

$4x \Rightarrow 1$ $\quad 2x \Rightarrow 1, \; x \Rightarrow \frac{1}{2}$

$x \Rightarrow \frac{1}{4}$ and $\frac{1}{2}$

Ex) Solve for x : $\frac{\log_2(9-2^x)}{x-3} \Rightarrow 1$

$\Rightarrow \log_2(9-2^x) \Rightarrow x-3$

$\rightarrow 9-2^x \Rightarrow 2^{x-3}$

$\rightarrow 9-2^x \Rightarrow \frac{2^x}{2^3}$

$\rightarrow 2^x \Rightarrow t$

$\rightarrow 9-t \Rightarrow \frac{t}{8}$

$\rightarrow \frac{t}{8}+t \Rightarrow 9$

$\rightarrow \frac{9t}{8} \Rightarrow 9$

$\rightarrow t \Rightarrow \frac{72}{9} \Rightarrow 8$

$\rightarrow 2^x \Rightarrow 8$

$\rightarrow 2^x \Rightarrow 2^3$

$\rightarrow x \Rightarrow 3$

"x" cannot be '3' cause the denominator cannot be '0'.

no. of Sol^n $\Rightarrow 0$

Ex) Solve for x:

$\log_{x-1} 4 = 1 + \log_2 x-1$

$\rightarrow \log_{x-1} 4 = 1 + \log_2 x-1$

$\{\log_a b = \log_c b / \log_c a\}$

$\rightarrow \frac{\log_2 4}{\log_2 (x-1)} \Leftrightarrow 1 + \log_2 (x-1)$

$\Rightarrow 2 \Rightarrow \log_2 (x-1) + (\log_2 x-1)^2$

$\Rightarrow \log_2 (x-1) \Leftrightarrow t$

$\rightarrow 2 \Rightarrow t^2 + t$

$\rightarrow t^2 + t - 2 \Rightarrow 0$

$\rightarrow t^2 + 2t - t - 2 \Rightarrow 0$

$\rightarrow t(t+2) - 1(t+2) \Rightarrow 0$

$\rightarrow (t+2)(t-1) \Rightarrow 0$

$\rightarrow t \Rightarrow -2$ and 1

$\rightarrow t \Rightarrow \log_2 x-1$

$\log_2 x-1 \Rightarrow -2$

$x-1 \Rightarrow \frac{1}{4}$

$x \Rightarrow 1 + 1/4$

$\therefore x \Rightarrow \frac{5}{4}$

$\log_2 (x-1) \Rightarrow 1$

$\rightarrow x - 1 \Rightarrow 2$

$x \Rightarrow 3$

Ex) Solve for x:

$$5^{1+\log_4 x} + 5^{(\log_{1/4} x)^{-1}} = \frac{26}{5}$$

$$\Rightarrow 5^{1+\log_4 x} + 5^{(\log_{1/4} x)^{-1}} = \log \frac{26}{5}$$

$$\rightarrow 5^{1+\log_4 x} + 5^{1/\log_{1/4} x} = \frac{26}{5}$$

$$\left\{ \frac{1}{\log_{1/4} x} \Rightarrow \log_x \frac{1}{4} \right\}$$

$$\log_a b \Rightarrow \frac{1}{\log_b a}$$

$$\Rightarrow 5^{1+\log_4 x} + 5^{\log_x \frac{1}{4}} = \frac{26}{5}$$

$$\Rightarrow 5\left[5^{(1+\log_4 x)} + 5^{-\log_x 4}\right] = 26$$

$$\Rightarrow 25^{1+\log_4 x} + 25^{-\frac{1}{\log_4 x}} = 26$$

$$\Rightarrow \log_4 x = t$$

$$\Rightarrow 25^{1+t} + 25^{-\frac{1}{t}} = 26$$

Ex) Value of :-

$$\left[\left[\log_2 9\right]^2\right]^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}} \text{ is}$$

$$\Rightarrow \left[\left[\log_2 9\right]^2\right]^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}}$$

$$\Rightarrow \because \sqrt{7}^{\log_4 7} \Rightarrow \sqrt{7}^{\log_7 4} \Rightarrow 4^{\log_7 7^{1/2}}$$

$$\Rightarrow 4^{\frac{1}{2}\log_7 7} \to 1$$

$$\Rightarrow (4)^{\frac{1}{2}} \Rightarrow (2)^{2\times\frac{1}{2}}$$

$$\Rightarrow \underline{\underline{2}}$$

$$\Rightarrow \left[\left[\log_2 9\right]^2\right]^{\frac{1}{\log_2(\log_2 9)}}$$

$$\Rightarrow \log_2 9 \Rightarrow t$$

$$\to \left((t)^2\right)^{\frac{1}{\log_2 t}} \Rightarrow \left(t^2\right)^{\log_t 2} \Rightarrow (t)^{\log_t 2 \times \log_t 2}$$

$$\Rightarrow 2^{\log_t t^1 \times \log_t t^1}$$

$$\Rightarrow 2\times 2 \Rightarrow 4$$

$$\to \left[\left[\log_2 9\right]^2\right]^{\frac{1}{\log_2(\log_2 9)}} \times \sqrt{7}^{\frac{1}{\log_4 7}} \Rightarrow 2\times 2\times 2 \Rightarrow \underline{\underline{8}}$$

Ex) If $3^x = 4^{x-1}$ then $x = ?$

(A) $\frac{2\log_3 2}{2\log_3 2 - 1}$ (B) $\frac{2}{2 - \log_2 3}$ (C) $\frac{1}{1 - \log_4 3}$

(D) $\frac{2\log_2 3}{2\log_2 3 - 1}$

$\Rightarrow 3^x = 4^{x-1}$

Opt: A (Not possible)

$2\log_3 2 - 1 \Rightarrow 0$

$\neq 0$

Opt: B

$3^{\frac{2}{2-\log_2 3}} \Leftrightarrow 4^{\frac{2}{2-\log_2 3}-1}$

(C) Not

(c) $\frac{1}{1-\log_4 3}$

Solve

i) $\log_{x-1} 3 \Rightarrow 2$

ii) $\log_4(2\log_3(1+\log_2(1+2\log_3 x))) \Rightarrow \frac{1}{2}$

iii) $\log_3(1+\log_3(2^x-7)) \Rightarrow 1$

iv) $\log_3(3^x-8) \Rightarrow 2-x$

v) $\log_2(9-2^x) / 3-x \Rightarrow 1$

vi) $\log_{5-x}(x^2-2x+65) \Rightarrow 2$

vii) $\log_{10} 5 + \log_{10}(x+10) - 1 \Rightarrow \log_{10}(21x-20) - \log_{10}(2x-1)$

viii) $x^{1+\log_{10} x} \Rightarrow 10x$

ix) $2(\log_x \sqrt{5})^2 - 3\log_x \sqrt{5} + 1 \Rightarrow 0$

x) $3+2\log_{x+1} 3 \Rightarrow 2\log_3(x+1)$

Logrithmic Inequality

Concept: Log Wali Inequality

$\log_2 x > 1 \Rightarrow x > 2^1$

∴ Base > 1 ⇒ sign of inequality = same

∴ Base < 1 ⇒ sign of inequality = change

E.g $\log_{1/2} x < 3$

$x < \left(\frac{1}{2}\right)^3$

$x > \frac{1}{8}$

ii) $\log_3(x-1) > 2$

$x - 1 > (3)^2$

$x - 1 > 9$

$x > 10$

Domain of Log function

Concept:

$f(x) = \log_{10}(x-1)$

this box should be greater than 0

- $\log_a x$
  - $x > 0$ $\{x \in (0, \infty)$
  - $a > 0$
  - $a \neq 1$

$x - 1 > 0$ {To find the domain}

$x > 1$

$x \in (1, \infty)$

1 is not included.

Eg) $f(x) = \log_2(3-x)$

$3 - x > 0$

$-x > -3$

$x < 3$

~~$x \in [0,1,2]$~~

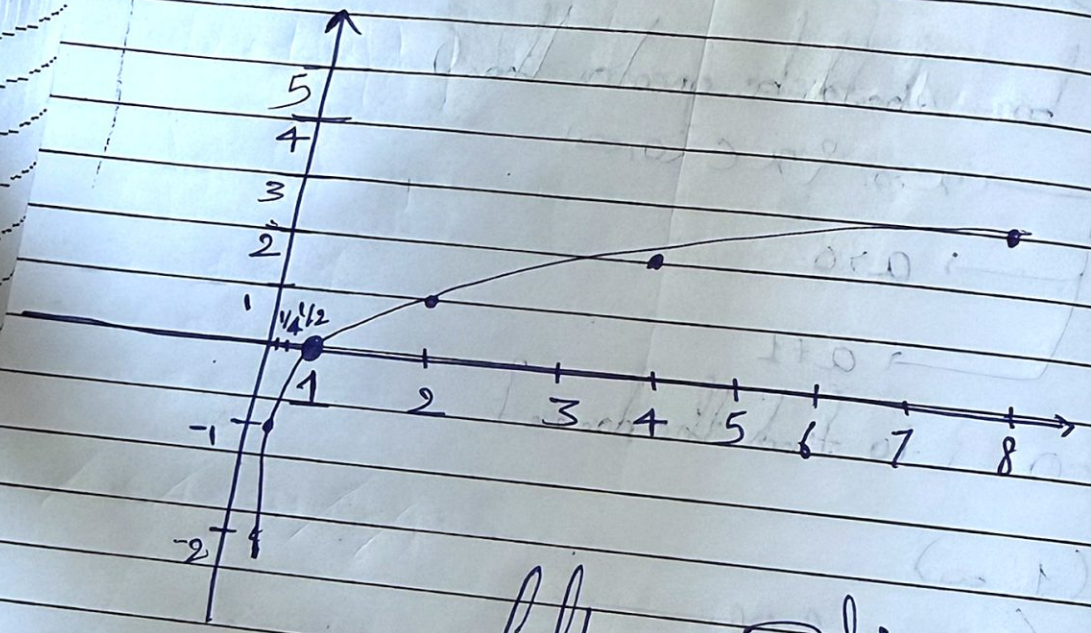
~~$x \in [0,3)$~~

$\Rightarrow x \in (-\infty, 3)$

Graph of Log function

∴ $y = \log_2 x$

∴ When Base > 1



Left → Right

∴ Increasing function

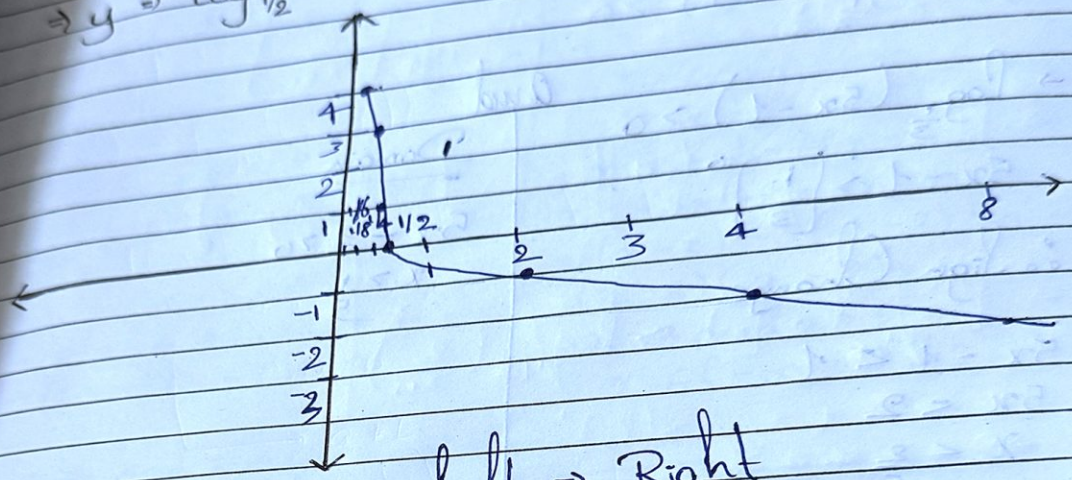
∴ $y = \log_2 x$

$x = 2^y$

x	y
1	0
2	1
4	2
8	3
1/2	-1
1/4	-2

∴ When Base < 1

$\Rightarrow y = \log_{1/2} x$



Left → Right

Decreasing

$\rightarrow y = \log_{1/2} x$

$\rightarrow x \Leftrightarrow \left[\frac{1}{2}\right]^y$

x	y
1	0
$\frac{1}{2}$	1
$\frac{1}{4}$	2
$\frac{1}{8}$	3
$\frac{1}{16}$	4
2	-1
4	-2
8	-3

Questions

Ex) $\log_{\frac{1}{3}}(5x-1) > 0$

$\rightarrow \log_{\frac{1}{3}}(5x-1) > 0$

$5x-1 > \left(\frac{1}{3}\right)^0$

$\therefore$ Sign Change

$5x-1 < 1$

$5x < 2$

$x < \frac{2}{5}$

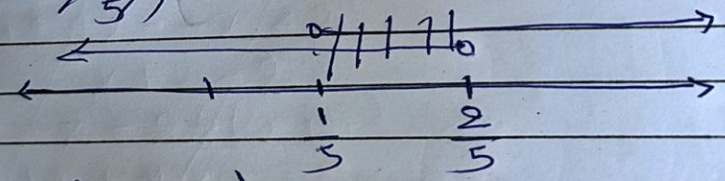
And

Domain

$5x-1 > 0$

$x > \frac{1}{5}$

$x \in \left(-\infty, \frac{2}{5}\right)$



$x \in \left(\frac{1}{5}, \frac{2}{5}\right)$

Ex) $\log_{0.5}(x^2-5x+6) > -1$

$\Rightarrow \log_{\frac{1}{2}}(x^2-5x+6) > -1$

$\therefore$ Sign change

$\rightarrow x^2-5x+6 < 2$

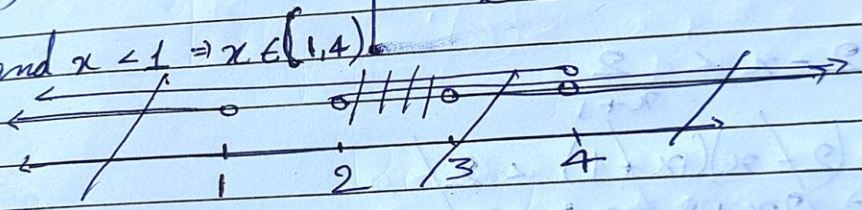
$\rightarrow x^2-5x+4 < 0$

$\rightarrow x^2-4x-x+4 < 0$

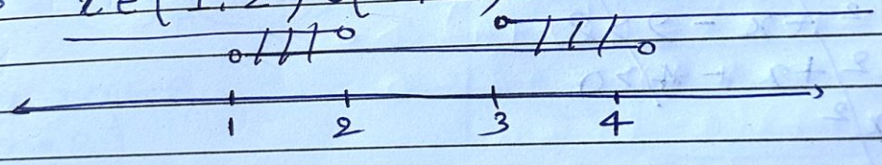
$\rightarrow x(x-4)-1(x-4) < 0$

$\rightarrow (x-4)(x-1) < 0$

$\rightarrow x < 4$ and $x < 1 \Rightarrow x \in (1,4)$



$\rightarrow x \in (1,2) \cup (3,4)$



Domain

$x^2-5x+6 > 0$

~~$x(x-6)+x+6$~~

~~$x(x-6)+1(x-6)$~~

~~$x(x+1)-6(x+1)$~~

$x^2-3x-2x+6 > 0$

$x(x-3)-2(x-3) > 0$

$(x-3)(x-2) > 0$

$x > 3$ and $x > 2$

$x \in (-\infty, 2) \cup (3, \infty)$

Concept

⇒ If $x_1 > x_2$ and You add log on both sides and if the base is greater than "1" the sign remains same and if base is less than "1" the sign will change.

⇒ If $\log_2 x_1 > \log_2 x_2$ where base is greater than 1 and if you cancel out the log on both side, since the base is greater than 1 the sign remains same. If it was

less than 1 [i.e base] the sign of inequality will change.

Ex) $\log_{\frac{1}{4}}(2-x) > \log_{1/4}\left[\frac{2}{x+1}\right]$

→ Since base is less than 1 the sign of inequality will change

→ $\log_{1/4}(2-x) > \log_{1/4}\left(\frac{2}{x+1}\right)$

→ $2-x < \frac{2}{x+1}$

→ $(2-x)(x+1) < 2$

→ $2x+2-x^2-x < 2$

→ $-x^2-x+2 < 2$

→ $x^2+x-2 > 2$

→ $x^2+x-4 > 0$

→ x^2

→ $2-x < \frac{2}{x+1}$

→ $2-x-\frac{2}{x+1} < 0$

→ $\frac{2(x+1)-x(x+1)-2}{x+1} < 0$